

AI Architecture and Design

Course Assignment

User Interface Using Streamlit

1 Assignment Overview

Streamlit is a Python framework used to build interactive web applications, especially for data science, machine learning, and AI projects, without requiring much knowledge of HTML, CSS, or JavaScript.

GitHub is an online platform where developers can store their code, track changes, collaborate with others, and share software projects.

In this assignment, you will improve the web interface by adding more hyperparameters for all the machine learning model available on the interface. In addition, you should push all changes made to your code to GitHub.

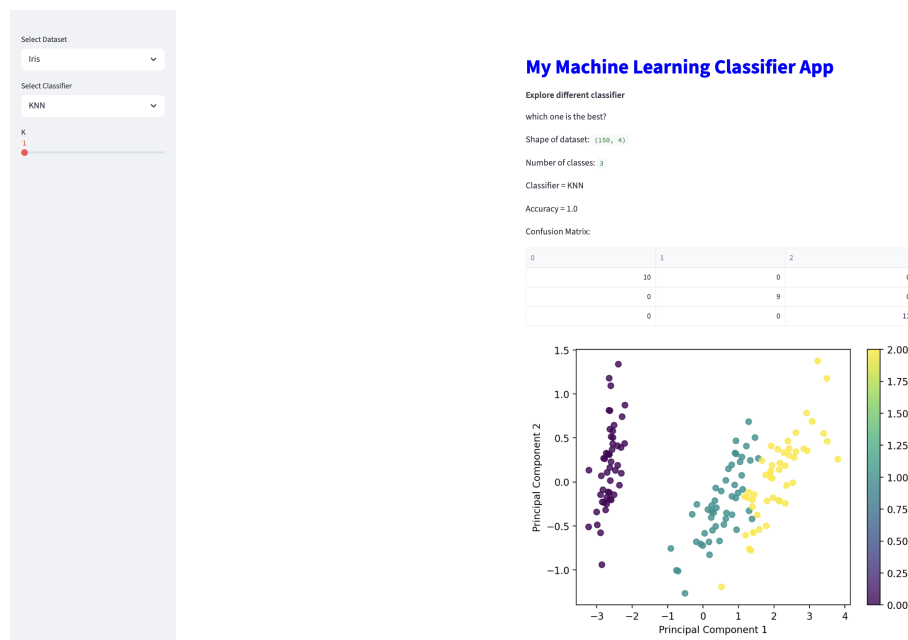


Figure 1: Screenshot of the Web Interface

2 Learning Objectives

After completing this project, you should be able to:

- Create a web application interface using Streamlit.

- Store your code, track changes, collaborate with others, and share software projects..

3 Part I: Adding Hyperparameters to the Machine Learning Models

Add an extra hyperparameters to the provided code:

- Add hyperparamter "leaf_size" to the K-Nearest Neighbors(KNN) model.
<https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>
- Add hyperparamter "degree" to the Support Vector Classifier(SVC) model.
<https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html>
- Add hyperparamter "min_samples_split" to the Random Forest(RF) model.
<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>

4 Final Deliverables

The assignments consists of the following deliverables.

- Submit a full screenshot of the new web interface showing all the newly added hyperparameters.
- Submit a full screenshot of your GitHub page showing your updated code.
- Submit your Python file.